F79SP Session 15: Further theories on stationary distributions and limiting distributions(2)

Case 3: S is finite, there are ≥ 2 closed classes and each of them is aperiodic

- Let $C_1, C_2, \ldots, C_r$ be all the closed classes of the MC, and they are all aperiodic.
- Each C_k forms a MC. We can obtain a unique stationary distribution $\boldsymbol{\pi}^{C_k}$ within. Then, construct:

$$\boldsymbol{\pi}^{(k)} = \begin{cases} \pi_i^{C_k}, & i \in C_k \\ 0, & \text{otherwise} \end{cases}$$

  Such $\boldsymbol{\pi}^{(k)}$ is a stationary distribution of the MC.

- Hence, stationary distributions of the MC are non-unique, all in the form of linear combination of $\boldsymbol{\pi}^{(k)}$'s as follows:

$$\boldsymbol{\pi} = \sum_{k=1}^{r} \gamma_k \boldsymbol{\pi}^{(k)},$$

  where each $\gamma_k \geq 0$, $\sum_{k=1}^{r} \gamma_k = 1$.

- For any initial distribution, the associated limiting distribution $\boldsymbol{\omega}$ exists, and is dependent on the initial distribution. More details are as follows:

  1. If $X_0 = i$ for $i \in C_k$ for some k (start from a recurrent state), then $\boldsymbol{\omega} = \boldsymbol{\pi}^{(k)}$.
  2. If $X_0 = i$ and i does not belong to any C_k (start from a transient state), then:

$$\boldsymbol{\omega} = \sum_{k=1}^{r} \alpha_i^{C_k} \boldsymbol{\pi}^{(k)},$$

     where $\alpha_i^{C_k}$ is the absorption probability of state i into closed class C_k.
  3. If X_0 is random, then we have the following result:

$$\boldsymbol{\omega} = \sum_{i \in S} \left(P(X_0 = i) \sum_{k=1}^{r} \alpha_i^{C_k} \boldsymbol{\pi}^{(k)} \right),$$

Example 0.1. *Consider a MC with transition matrix* $P = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{6} & \frac{1}{6} & 0 & 0 & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{4} & 0 & 0 & 0 & \frac{1}{4} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$.

1. *Find the general form of stationary distributions of the MC.*
2. *Determine the limiting distribution given that* $X_0 = 7$.
3. *Determine the limiting distribution given that* $X_0 = 1$.
4. *Determine the limiting distribution given that* $X_0 = 5$.
5. *Determine the limiting distribution given that* X_0 *takes value either* 1 *or* 5*, each with probability* $\frac{1}{2}$.

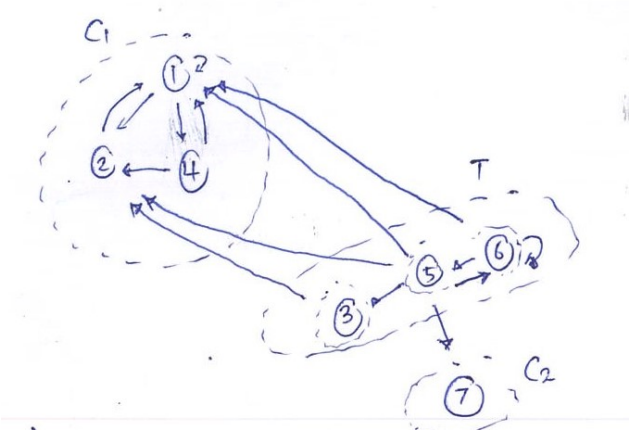


Figure 1: State space diagram of Example.

TBC

Finally, we revisit the proof of following result.

Theorem 4.4.3 and 4.4.4 in notes: Suppose that a THMC is such that S **is a single closed class and is aperiodic**, then:

1. There exists a vector (not necessarily a valid distribution) $\boldsymbol{\omega} = (\omega_j)_{j\in S}$ such that for all $i, j \in S$, $\lim_{n\to\infty} p_{ij}^{(n)} = \omega_j$ (independent of i).
2. In addition, if S **is finite**, then $\boldsymbol{\omega}$ above is the unique stationary distribution.

Proof. We will only prove 2. by assuming 1. is true. The proof consists of three parts, to show $\boldsymbol{\omega}$ is a distribution, $\boldsymbol{\omega}$ is stationary, and its uniqueness.

Part 1: $\boldsymbol{\omega}$ is a distribution

- If S is finite, we have:

$$\begin{aligned}\sum_{j\in S} \omega_j &= \sum_{j\in S} \lim_{n\to\infty} p_{ij}^{(n)} \\ &= \lim_{n\to\infty} \sum_{j\in S} p_{ij}^{(n)} \text{ (Since } S \text{ is finite)} \\ &= \lim_{n\to\infty} 1 = 1.\end{aligned}$$

Part 2: $\boldsymbol{\omega}$ is stationary

- We have from Chapman-Kolmogorov equations, for any $n \geq 1$ and $i, j \in S$:

$$p_{ij}^{(n+1)} = \sum_{k\in S} p_{ik}^{(n)} p_{kj}.$$

- Then we show the stationarity conditions $\omega_j = \sum_{k\in S} \omega_k p_{kj}$ by letting $n \to \infty$:

$$\begin{aligned}\lim_{n\to\infty} p_{ij}^{(n+1)} &= \lim_{n\to\infty} \sum_{k\in S} p_{ik}^{(n)} p_{kj} \\ \omega_j &= \sum_{k\in S} \lim_{n\to\infty} p_{ik}^{(n)} p_{kj} \text{ (Since } S \text{ is finite)} \\ \omega_j &= \sum_{k\in S} \omega_k p_{kj}\end{aligned}$$

Part 3: Uniqueness of $\boldsymbol{\omega}$

- We suppose $\boldsymbol{\pi}$ is any stationary distribution of MC and show that it must be that $\boldsymbol{\pi} = \boldsymbol{\omega}$, or equivalently $\pi_j = \omega_j$ for every $j \in S$.
- Indeed, since $\boldsymbol{\pi} P^n = \boldsymbol{\pi}$, we have the following:

$$\begin{aligned}\pi_j &= \sum_{k\in S} \pi_k p_{kj}^{(n)} \\ \lim_{n\to\infty} \pi_j &= \lim_{n\to\infty} \sum_{k\in S} \pi_k p_{kj}^{(n)} \\ \pi_j &= \lim_{n\to\infty} \sum_{k\in S} \pi_k p_{kj}^{(n)} \text{ } (\pi_j \text{ is constant of } n) \\ \pi_j &= \sum_{k\in S} \lim_{n\to\infty} \pi_k p_{kj}^{(n)} \text{ (Since } S \text{ is finite)} \\ \pi_j &= \sum_{k\in S} \pi_k \lim_{n\to\infty} p_{kj}^{(n)} \\ \pi_j &= \sum_{k\in S} \pi_k \omega_j = 1(\omega_j) = \omega_j\end{aligned}$$

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